

TECHNICAL PROGRESS REPORT

Source-Linked Compression-Set Data for Silicone Formulations

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Executive summary

This report summarizes the work completed to assemble a source-linked record of silicone formulations and compression-set measurements from patents and journal articles. The practical question is whether a documented formulation and its test conditions can support useful guidance for seal design without requiring a new compression-set experiment for every candidate.

The main result is a funnel. We began with 2,925 downloaded documents. 1,930 contained at least one extracted property, 183 reported a compression-set value, and only 85 provided a value together with a composition and the three essential test conditions: temperature, time, and deflection. After source checking and curation, the current release contains 1,054 measurements over 605 formulations from 67 documents. The decrease is not a failure of the project; it identifies the amount of the published record that is sufficiently described to support quantitative comparison.

The release also distinguishes the material classes that are important for interpreting the data. A formulation is counted as a composite when it contains a dispersed filler role, and as a foam when it contains a blowing agent role. There are 525 composite formulations and 115 foam formulations. 110 formulations belong to both groups; the remaining formulations are 415 composite only, 5 foam only, and 75 neither.

The work produced a reliable baseline and a clear limitation. Test conditions alone give a leave-one-family-out R^2 of 0.289 with a mean absolute error of 17.5 percentage points. The chemistry representations examined so far do not produce a statistically separable improvement when entire source families are held out. This does not mean that chemistry is unimportant. It means that the present record is still too small, too unevenly described, and too concentrated in a limited number of source families to establish a transferable chemistry effect.

1 What are the major goals of the project?

Compression set is a useful engineering property because it measures how much of an imposed deformation remains after an elastomer has been held under load for a specified time and temperature [1]. It is directly related to the long-term performance of seals, but it is also strongly dependent on the test protocol. A value measured after 22 h at 175 °C is not directly interchangeable with a value measured after 70 h at 150 °C.

The first goal is therefore to gather measurements together with the conditions that make them interpretable. The second is to preserve the chemistry of the formulation: base polymer, crosslinker, catalyst or initiator, fillers, additives, and any blowing agent. Silicone elastomers are not single-molecule materials. Their behavior depends on the network created by the cure chemistry, the interaction with fillers, and the processing and aging history [2–4].

The third goal is to determine what can reasonably be learned from the published record. Polymer formulations are difficult to model because the design space is large while much of the practical knowledge is recorded in qualitative descriptions, patents, and tables with inconsistent terminology [5–7]. The project therefore asks two separate questions: what information can be recovered and checked, and how much predictive value does the recovered chemistry add beyond the test conditions themselves?

2 What was accomplished under these goals?

The project established a complete path from a broad literature record to a curated set of comparable measurements. Documents were collected from patents and journal literature, screened for silicone materials and quantitative compression-set results, and read in the context of the surrounding formulation and testing descriptions. Duplicate source families were treated as one scientific source so that repeated publication of the same work did not create artificial evidence.

The result is summarized in Figure 1. The top panel makes the funnel explicit: each additional requirement removes documents that do not provide enough information for a defensible comparison. The lower-right panel records the material classes in the final release. These classes overlap by design, because a filled silicone foam is both a composite and a foam.

2.1 Major Activities

The work had four connected activities. First, the source record was expanded without requiring the property to appear in a title or abstract. This was important because many patents present compression-set values only in example tables. Second, formulations were read as recipes rather than as isolated chemical names. Trade names, component labels, filler descriptions, cure systems, and amount conventions were retained when they were part of the source’s description.

Third, test conditions were connected to the measurements to which they belong. Temperature, time, deflection, medium, specimen geometry, and stated test standard were kept separate because each can change how a compression-set value should be interpreted. When a source stated conditions in a methods paragraph rather than beside the result, the conditions were connected back to the relevant measurements only when the source supported that connection.

Fourth, the resulting records were checked against the source pages. A stratified audit found the released compression-set value correct in 96.8% of checkable cases, with a Wilson 95% interval

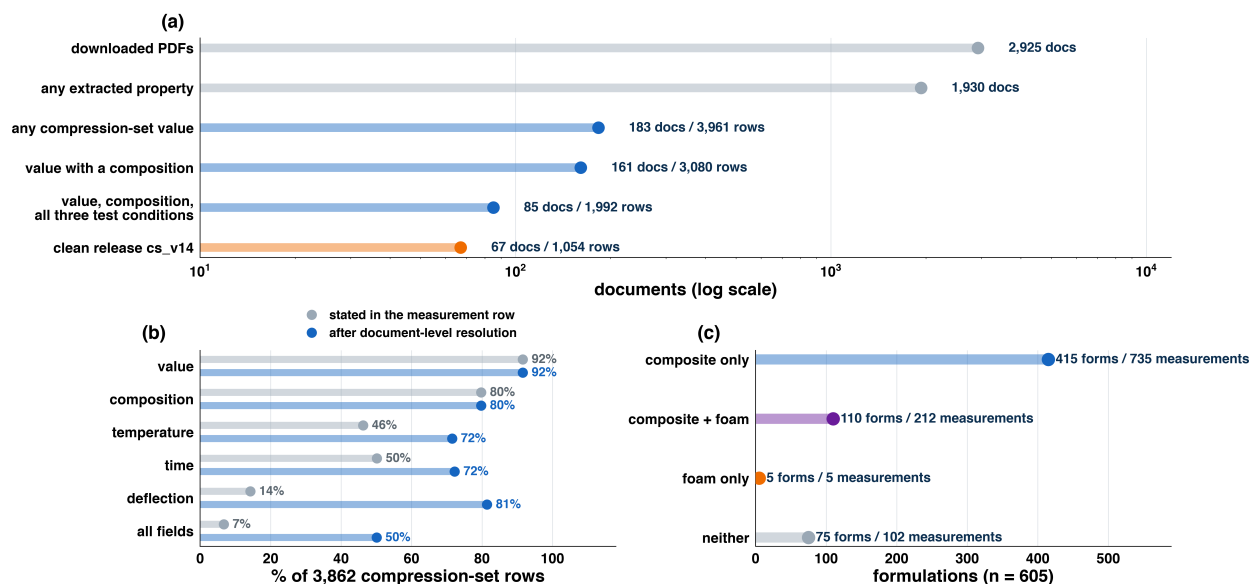


Figure 1: The record narrows from documents to usable measurements. (a) Documents remaining as composition, test protocol, and release-quality requirements are applied; row counts are shown where the stage is defined on measurements. (b) Information stated in the measurement row itself compared with information recovered from the document as a whole. (c) Formulations classified as composite, foam, both, or neither. Composite means a dispersed-filler role; foam means a blowing-agent role.

of 94.7–98.1%. Temperature and deflection were correct in every checked case. The audit also identified specific table-reading failures and source-family duplicates that would have been difficult to detect from numerical consistency alone.

2.2 Specific Objectives

Objective 1 – Build a quantitative source record. The project converted a dispersed set of patents and articles into a collection in which each measurement remains connected to its source document, formulation, and test description.

Objective 2 – Distinguish material classes. The project now reports which released formulations contain dispersed fillers and which contain blowing agents. This makes the composite and foam populations visible instead of combining them into one undifferentiated silicone category.

Objective 3 – Establish a defensible predictive baseline. The project compared condition-based prediction with several broad descriptions of chemistry while keeping source families separate during evaluation. The purpose was to identify what the current evidence supports, not to maximize a single score by mixing related examples between training and testing.

3 Significant Results

3.1 The document funnel identifies the quantitative core

The funnel begins with 2,925 downloaded documents and ends with 67 source documents in the current release. In between, the record passes through 1,930 documents with any extracted property, 183 documents with a compression-set value, 161 documents with a stated composition, and 85 documents with a complete test protocol. The corresponding compression-set row counts are 3,961 for the broad property population, 3,080 with a composition, and 1,992 with all three conditions.

This sequence is the most important result for planning future work. A large number of documents does not automatically create a large modeling dataset. The limiting resource is the number of source families that report a complete recipe, a measured value, and enough test information for comparison. The funnel shows exactly where additional reading and curation can increase the usable population.

3.2 The release separates composites and foams

The material classification is based on the ingredient roles supported by the source. Composite means at least one reinforcing, functional, extending, or lightweight filler. Foam means a blowing agent. The four categories below partition the 605 released formulations, while the aggregate composite and foam populations overlap.

Table 1: Operational material classes in release cs_v14. The composite and foam labels can overlap; the four rows partition the formulations.

Formulation class	Formulations	Compression-set measurements
Composite only	415	735
Composite and foam	110	212
Foam only	5	5
Neither composite nor foam	75	102
Total	605	1,054

In aggregate, 525 formulations from 60 documents are composites and 115 formulations from 17 documents are foams. The overlap is substantive rather than an accounting error: the 110 formulations in both classes contribute to both totals. This distinction should be carried into future comparisons because foam structure, filler content, and cure conditions can affect compression set in different ways.

3.3 The current release is broad but uneven

The current release cs_v14 contains 1,054 measurements over 605 formulations from 67 documents, including 65 patents and 2 journal articles published between 1974 and 2026. The median compression set is 25% and the interquartile range is 13–51%. The sources use several amount conventions and describe cure systems, fillers, and test geometries with different levels of detail.

Figure 2 summarizes this variation. The measurements are unevenly distributed across source documents, and the test conditions cluster around common industrial protocols while extending to

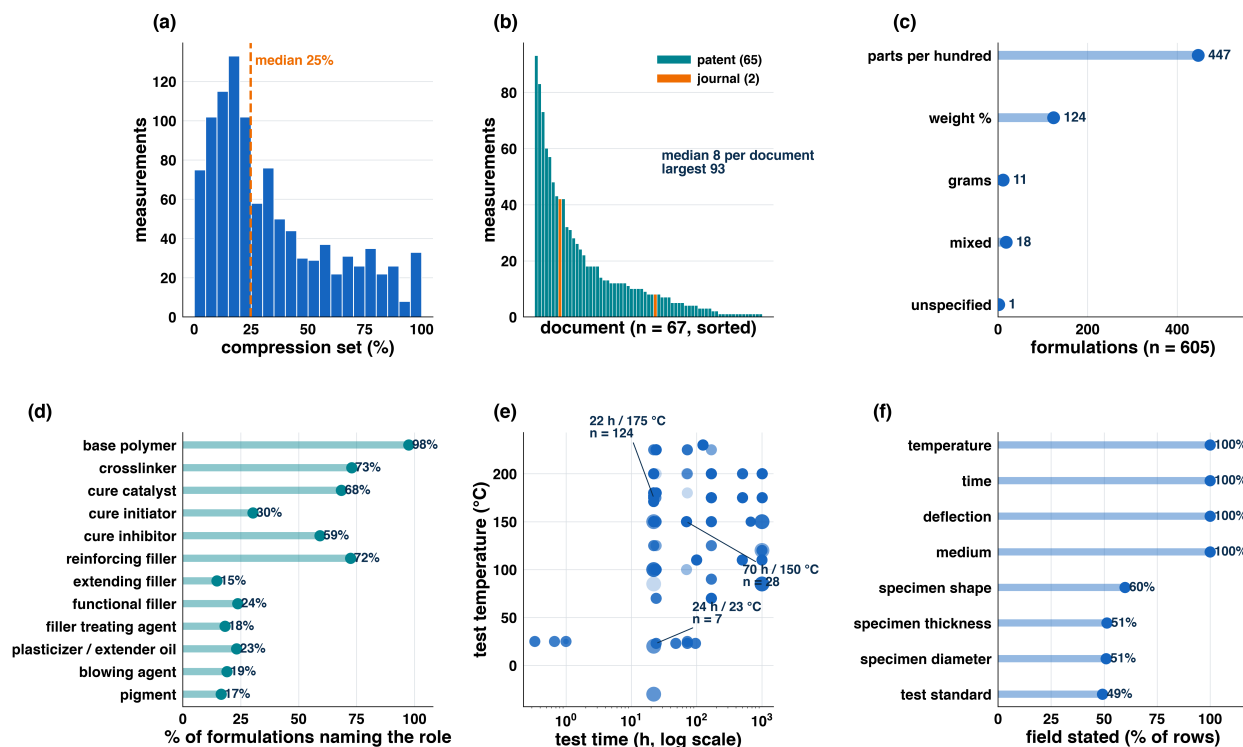


Figure 2: What the current release contains. The release includes measurements spanning a wide range of compression-set values, source documents, formulation amount conventions, ingredient roles, and test protocols. Empty fields are retained as evidence that the source was silent; they are not silently filled by an assumed value.

other temperatures and times. This unevenness is why the report emphasizes source families and document-level comparison rather than treating every row as an independent piece of evidence.

3.4 A conditions-only baseline is presently the defensible result

The first predictive comparison used the test conditions as the common basis for every formulation. Additional chemistry descriptions were then tested to ask whether they improved transfer to source families that had not been used to fit the prediction. Conditions alone reached a leave-one-family-out R^2 of 0.289 with a mean absolute error of 17.5 percentage points.

Adding stated viscosity, ingredient descriptions, combined descriptors, or ingredient categories changed the leave-one-family-out score by small amounts whose uncertainty intervals included zero. The best chemistry representation reached 0.280, but the current corpus cannot establish that this is better than the conditions-only baseline. Random-row evaluation gives a substantially higher diagnostic score, including 0.737 for ingredient categories, because related examples from the same document remain on both sides of the comparison.

Figure 3 presents this result as a model ladder. The main conclusion is a statement about evidence: the present release supports a useful condition-based baseline, but it does not yet support a claim that the available chemistry descriptions improve prediction for unseen source families. A larger and more balanced set of complete source families, or a trusted validation set, is needed to answer that question.

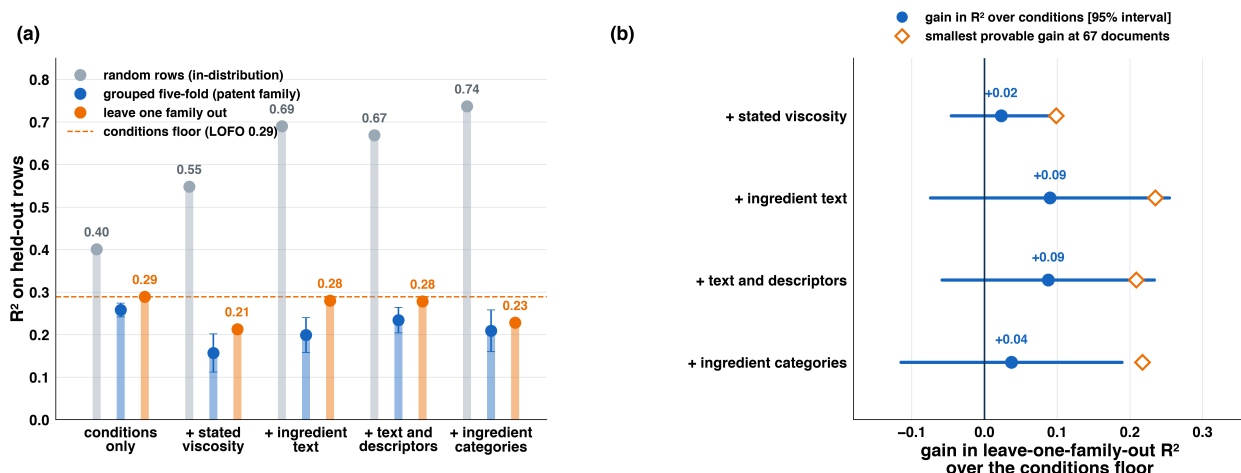


Figure 3: The predictive baseline and the chemistry question. The conditions-only baseline is compared with several broad chemistry descriptions under familiar-document and unseen-family evaluations. The chemistry gains remain within the uncertainty supported by the current number of source families.

4 Lessons from verification

Several verification cases changed the way the record is interpreted. A correct numerical value can still be attached to the wrong formulation when two members of a patent family present different tables. A plausible chemical name can still be wrong when a source uses a similar-looking end group. A table can appear complete while a shifted column has assigned an amount or a test condition to the neighboring row. These cases show why numerical agreement, chemical plausibility, and direct source checking are all needed.

The most consequential lesson is that source context matters more than the title of a document. Some documents that appear general from their title contain the most useful aging data, while some documents that mention silicone do not describe a silicone matrix and should not be included. The funnel and the audit are therefore complementary: the funnel shows how many documents remain, while the audit tests whether the surviving values still mean what the source reported.

5 Companion benchmark and broader value

The table-reading work was also tested on a 1968 reference volume containing tabulated organosilicon properties. That setting offered a fixed ground truth against which names, structures, values, and pressures could be checked. The verified benchmark contains 509 measurements over 436 structures from 159 references. The benchmark was useful because it separated errors in reading a table from uncertainty in interpreting a modern patent formulation.

The broader contribution is a change in the unit of analysis. Silicone formulation development is often discussed in terms of individual ingredients or isolated property values. The present record connects the source document, the complete formulation, the material class, and the test protocol. That connection makes it possible to ask which evidence is genuinely comparable and which apparent trend is only a consequence of how a document was written.

6 How have the results been disseminated?

The results are being organized as a technical report, a frozen research release, source-linked records for the collaborating group, and a documented set of figures that communicate both the size of the record and its limits. The funnel is intended to make acquisition priorities visible to collaborators: new documents are valuable when they add complete formulations and complete test conditions, not merely when they increase the number of downloaded papers.

The report also provides a common language for discussing material classes. Future meetings can distinguish filled silicone composites, silicone foams, and formulations that belong to both categories instead of comparing all of them as one population.

7 What do you plan to do next to accomplish the goals?

The next stage has three priorities. First, confirmed errors from the audit will be corrected and additional source families will be added through the same source-linked process. Second, the composite and foam labels will be reviewed alongside cure system and base-polymer information so that the material classes can be compared without conflating structure and chemistry. Third, the predictive baseline will be reevaluated after the record is expanded and after collaborators provide a small validation set with trusted formulations and test conditions.

Two inputs from collaborators would have immediate value. A set of formulations that Los Alamos trusts, with or without measured values, would provide a fixed reference for evaluating extraction and representation choices. Also needed is a statement of what prediction error would change a design decision at a named test protocol. That engineering threshold would determine whether the next effort should focus on more source families, more complete chemistry descriptions, or targeted measurements of composites and foams.

8 Information retained in the research release

Each released measurement remains connected to its source document and to the formulation from which it was reported. The retained information includes the ingredient descriptions and roles, amounts and amount conventions, measured compression set, test temperature, time, deflection, and any stated medium, specimen geometry, or post-cure history. Missing information is left visible as missing so that future users can distinguish a source that did not report a quantity from a formulation that did not contain or require it.

The current release is a snapshot of the evidence available at this stage of the project. Its value is not only the number of rows; it is the traceability that allows a future correction, new material class, or new test condition to be connected back to the page on which the original claim was made.

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